

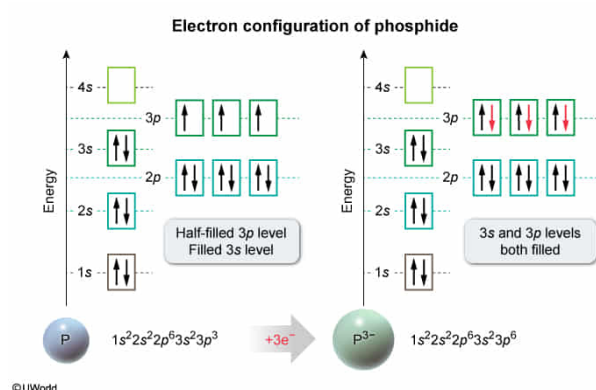
Which statement correctly describes both the 3s and 3p energy levels in the lowest-energy electron configuration of the phosphide anion P^{3-} ?

- ☐ A. Both the 3s and 3p levels are half-filled. [2%]
- ☐ B. The 3s level is filled and the 3p level is half-filled. [15%]
- ☐ C. The 3s level is half-filled and the 3p level is filled. [2%]
- ☒ D. Both the 3s and 3p levels are filled. [79%]

Correct

79% answered correctly

Explanation:



An **electron configuration** accounts for the placement of all electrons within the **shells** and **subshells** of an atom or ion. **Shells** are indicated using the principal quantum number n , which relates to the distance of an orbital from the nucleus. **Subshells** are labeled by type (s , p , d , or f), which relates to the **shapes** of the occupied orbital, which hold a maximum of 2, 6, 10, and 14 electrons, respectively. **Electron configurations** are listed sequentially in **order** of **increasing energy**.

Elemental atoms contain an equal number of protons and electrons, resulting in no net charge. If an elemental atom gains or loses electrons and an ion is formed, the charge of the ion is determined by how many electrons are gained or lost. If electrons are lost (cation formation), electrons in the **highest-energy filled** shell/subshell level are **removed first**. Likewise, if electrons are gained (anion formation), the **lowest-energy unfilled** shell/subshell level is **filled first**.

An elemental atom of phosphorus has a total of 15 electrons and 15 protons (no net charge) with an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^3$. Therefore, a phosphide anion is the result of an elemental phosphorus atom gaining 3 additional electrons within the half-filled 3p subshell. As a result, the phosphide anion has a total of 18 electrons with a configuration of $1s^2 2s^2 2p^6 3s^2 3p^6$. This gives the phosphide anion a -3 charge and filled 3s and 3p energy levels.

(Choices A and C) Lower energy shells/subshells must be fully filled before electrons can be added to higher energy shells/subshells. The 3s level must be filled before adding electrons to 3p.

(Choice B) This is the lowest-energy electron configuration of a *phosphorus atom* rather than a phosphide anion.

Educational objective:

An electron configuration accounts for all electrons held by an atom or ion, listed sequentially by shell and subshell in order of increasing energy. During cation formation, electrons are lost from the highest-energy filled shell/subshell first; during anion formation, electrons are added to the lowest-energy unfilled shell/subshell first.

Time spent:0 sec

Subject:

General Chemistry

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System:

Atomic Theory and Chemical Composition